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GitHub Link	https://github.com/saimounika1315/MLA0403---Deep-Learning

SIMATS ENGINEERING

CO2 - ASSESSMENT TOOL 2

Scenario-Based Questions

COURSE NAME: **Deep Learning** **COURSE CODE:** **MLA04**

COURSE OUTCOME COVERED

CO 2: Students will be able to apply supervised learning algorithms such as linear regression, classification models, decision trees, neural networks, support vector machines, and ensemble methods to solve real-world prediction and classification problems. They will gain the ability to select appropriate models, train them using datasets, and evaluate their performance. Students will also understand the strengths and limitations of different supervised learning approaches. (BTL-4)

CO Weightage: 15%

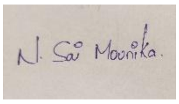
Assessment Tool Weightage: 40%

Duration: 90 minutes

Marks: 25

Code of Conduct:

I Narra Sai Mounika Reg no: 192425367 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Narra Sai Mounika

Signature of the student: **Name of the student:**

Scenario-Based Questions

CO-AT2

QUESTIONS: 05 **Total Marks:25**

1. **Unsupervised Training of Neural Networks, Auto Encoders**
   A hospital wants to identify abnormal ECG patterns from thousands of unlabeled records. Which deep learning technique from the syllabus would you recommend? Justify your choice and explain the workflow.
   Answer: For identifying abnormal ECG patterns from thousands of unlabeled records, I would recommend using an Autoencoder, an unsupervised deep learning technique.
   Justification:

- ECG data is unlabeled, making supervised learning difficult.
- Autoencoders learn the normal patterns present in the ECG signals.
- Abnormal ECG signals produce higher reconstruction errors, helping detect anomalies.

   Workflow:

1. Collect and preprocess ECG records.
2. Train the autoencoder using normal ECG patterns.
3. The encoder compresses the ECG data into a lower-dimensional representation.
4. The decoder reconstructs the original ECG signal.
5. Calculate reconstruction error between original and reconstructed signals.
6. Signals with high reconstruction errors are identified as abnormal.
7. Doctors can review the detected abnormal ECG patterns.

   Benefits:

- Works without labeled data.
- Detects hidden abnormalities efficiently.
- Reduces manual effort in analyzing ECG records.

2. **Auto Encoders, Representation Learning**
   A retail company wants to reduce the dimensionality of customer purchase data before performing customer segmentation. Explain how an autoencoder can be used to solve this problem.
   Answer: A retail company can use an Autoencoder to reduce the dimensionality of customer purchase data before customer segmentation.
   Process:

1. Input customer purchase records into the autoencoder.
2. The encoder compresses high-dimensional purchase data into a compact latent representation.
3. Important purchasing patterns are retained while redundant information is removed.
4. The compressed features are extracted from the bottleneck layer.
5. Clustering algorithms such as K-Means can then be applied to these features for customer segmentation.

Advantages:

- Reduces storage and computational requirements.
- Captures meaningful customer behavior patterns.
- Improves segmentation accuracy.
- Eliminates noise from the dataset.

Outcome:

The company can identify customer groups such as frequent buyers, seasonal shoppers, and premium customers more effectively.

3. **Width and Depth of Neural Networks, Activation Functions**

A facial recognition system performs poorly when trained using a shallow neural network. Recommend architectural changes involving network depth and activation functions.

Answer: A shallow neural network may fail to learn complex facial features required for accurate facial recognition.

Recommended Architectural Changes:

1. Increase Network Depth
    - Add more hidden layers.
    - Deep networks learn hierarchical features.
    - Early layers detect edges and textures.
    - Deeper layers recognize eyes, nose, and facial structures.
2. Optimize Network Width
    - Increase the number of neurons in hidden layers.
    - Allows learning of more feature combinations.
3. Use Better Activation Functions
    - Replace Sigmoid with ReLU (Rectified Linear Unit).
    - ReLU reduces vanishing gradient problems.
    - Faster convergence during training.
4. Advanced Activations
    - Leaky ReLU or ELU can improve learning performance.

Expected Result:

The deeper network can learn complex facial representations and improve recognition accuracy significantly.

4. **Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks**

A streaming service wants to recommend movies to users based on their viewing history without using labeled data. Explain how RBMs can be applied to build the recommendation engine.

Answer: A streaming service can use Restricted Boltzmann Machines (RBMs) to build a movie recommendation system using user viewing history.

Working Principle:

1. Create a user-movie interaction matrix.
2. Visible layer represents movies watched by users.
3. Hidden layer learns latent features such as:
   - Action movie preference
   - Comedy preference
   - Romance preference
4. Train the RBM using unsupervised learning.
5. The RBM learns relationships between users and movies.
6. Missing movie ratings or preferences are predicted.
7. Recommend movies with high predicted preference scores.

Advantages:

- No labeled data required.
- Learns hidden user interests automatically.
- Handles sparse viewing data effectively.

Outcome:

Users receive personalized movie recommendations based on learned viewing patterns.

5. **Deep Learning Applications, Representation Learning**

A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks.

Answer: To detect defective products from sensor readings collected every second, a deep learning solution combining Representation Learning and Neural Networks can be designed.

Proposed Solution:

1. Collect sensor data from manufacturing equipment.
2. Perform preprocessing and normalization.
3. Use an Autoencoder for representation learning.
4. The encoder extracts meaningful features from sensor readings.
5. These compressed features represent the health condition of products.
6. Feed the learned representations into a Deep Neural Network (DNN) classifier.
7. The classifier categorizes products as:
   - Normal
   - Defective

Workflow:

Sensor Data → Autoencoder → Feature Extraction → Deep Neural Network → Defect Detection

Advantages:

- Detects defects automatically in real time.
- Reduces manual inspection costs.
- Improves product quality and manufacturing efficiency.

- Learns complex sensor patterns effectively.
  Expected Outcome:
  The system can identify defective products early, reducing waste and improving production reliability.

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand